

6208NSC: Assignment 1: Enzyme Kinetics

This assignment is based on the Michaelis–Menten enzyme kinetics discussed during class. It is designed to consolidate the formulation and analysis of enzyme kinetic models.

Unless stated otherwise, assume that all reactions take place in a well-mixed system of constant volume, all rate constants are positive, and concentrations are non-negative. Use the notation adopted in class: a lower-case letter denotes the concentration of the corresponding upper-case chemical species. For example,

$$s = [S], \quad e = [E], \quad c = [C], \quad p = [P].$$

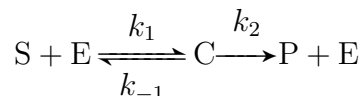
Apply this convention to other species and complexes; for example, $i = [I]$, $c_1 = [C_1]$ and $c_2 = [C_2]$. Clearly define any additional notation that you introduce.

Unless stated otherwise, assume that all enzyme is initially free, so that $e(0) = e_0 = e_T$, where e_T denotes the total enzyme concentration.

In this assignment, the reaction velocity means the net rate of product formation, $v = dp/dt$. For a reversible product-forming step, include both the forward and reverse contributions.

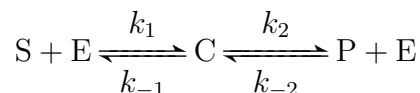
Submit a concise report showing clear working for each question. Include clearly labelled figures for numerical results and include the code used to generate them. Identify any software used. Marks are awarded for correct model formulation and derivation, appropriate use of assumptions, scientific interpretation, and clear presentation.

1. Consider the enzyme-catalysed reaction shown below.

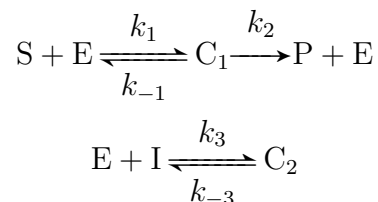


- (a) Derive the system of ordinary differential equations for s , e , c and p using the law of mass action. Identify the enzyme and substrate conservation relations.
- (b) Numerically solve the full ODE system from part (a). Choose plausible parameter values and initial conditions, and justify the choices briefly. Plot s , e , c and p against time and verify numerically that the conservation relations from part (a) are satisfied.
- (c) Using the rapid-equilibrium approximation, derive an analytical expression for the reaction velocity. State the assumption required. Use the dissociation constant $K_D = K_1 = k_{-1}/k_1$ and $V_{\max} = k_2 e_0$.
- (d) Using the quasi-steady-state approximation, derive an analytical expression for the reaction velocity. Use $K_M = (k_{-1} + k_2)/k_1$ and $V_{\max} = k_2 e_0$.
- (e) Select two parameter sets: one for which both approximations are expected to be valid and one for which at least one approximation is expected to be inaccurate. For each parameter set, plot $v = k_2 c$ from the full numerical solution against s . On the same axes, plot the rapid-equilibrium and quasi-steady-state velocity expressions using the same parameter values. Explain the principal differences and whether the assumptions underlying each approximation are satisfied.

2. In the previous question, the product-forming step was irreversible. We now allow product and free enzyme to recombine to form the enzyme–substrate complex, as shown below.

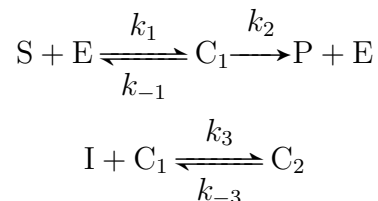


- Derive the system of ordinary differential equations for s , e , c and p using the law of mass action. Identify the conservation relations.
 - Using the quasi-steady-state approximation, derive an analytical expression for the net reaction velocity. State your assumptions and define any grouped parameters that you introduce.
 - Determine the limiting form of the velocity as $p \rightarrow 0$. Explain its physical meaning and its relationship to the quasi-steady-state result from Question 1.
 - Determine the limiting form of the velocity as $k_{-2} \rightarrow 0$. Explain how the reaction mechanism then relates to Question 1.
3. Consider the enzyme reaction shown below that includes a competitive inhibitor. In this reaction, $C_1 = ES$ and $C_2 = EI$; only C_1 can form product.

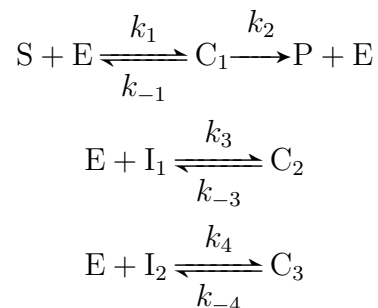


- Derive the system of ordinary differential equations using the law of mass action. Identify the total enzyme, total inhibitor and substrate-containing conservation relations.
- Using the quasi-steady-state approximation for both c_1 and c_2 , derive an analytical expression for the reaction velocity in terms of the free inhibitor concentration i . Assume that inhibitor is present in sufficient excess that i can be treated as approximately constant. Define the inhibition constant K_I , and explain how competitive inhibition changes the apparent K_M and $V_{\max}$.

4. Consider the enzyme reaction shown below that includes an uncompetitive inhibitor. Here, $C_1 = ES$ and $C_2 = ESI$; assume that C_2 is catalytically inactive and can dissociate only through the reaction shown.



- (a) Derive the system of ordinary differential equations using the law of mass action. Identify the total enzyme, total inhibitor and substrate-containing conservation relations.
- (b) Using the quasi-steady-state approximation for both c_1 and c_2 , derive an analytical expression for the reaction velocity in terms of the free inhibitor concentration i . Assume that inhibitor is present in sufficient excess that i can be treated as approximately constant. Define K_I , and explain how uncompetitive inhibition changes the apparent K_M and $V_{\max}$.
5. Consider the following enzyme-catalysed reaction that includes two different competitive inhibitors. Here, $C_1 = ES$, $C_2 = EI_1$ and $C_3 = EI_2$.



- (a) Derive the system of ordinary differential equations using the law of mass action. Identify the total enzyme, individual inhibitor and substrate-containing conservation relations.
- (b) Using the quasi-steady-state approximation for c_1 , c_2 and c_3 , derive an analytical expression for the reaction velocity in terms of the free inhibitor concentrations i_1 and i_2 . Assume that both inhibitors are present in sufficient excess that i_1 and i_2 can be treated as approximately constant. Define the two inhibition constants and show that your expression reduces to the single-inhibitor result when either inhibitor concentration is zero.